8、MoveIt轨迹规划

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8.1、启动
8.2、源码解析

本节课以MoveIT仿真为例，如果需要设置真机与仿真同步，请看【02、MoveIt注意事项与控制真机】这节课。！！！注意安全！！！

效果演示为虚拟机，其他主控运行情况（与主控性能有关，根据实际情况而定）。

8.1、启动

启动MoveIT

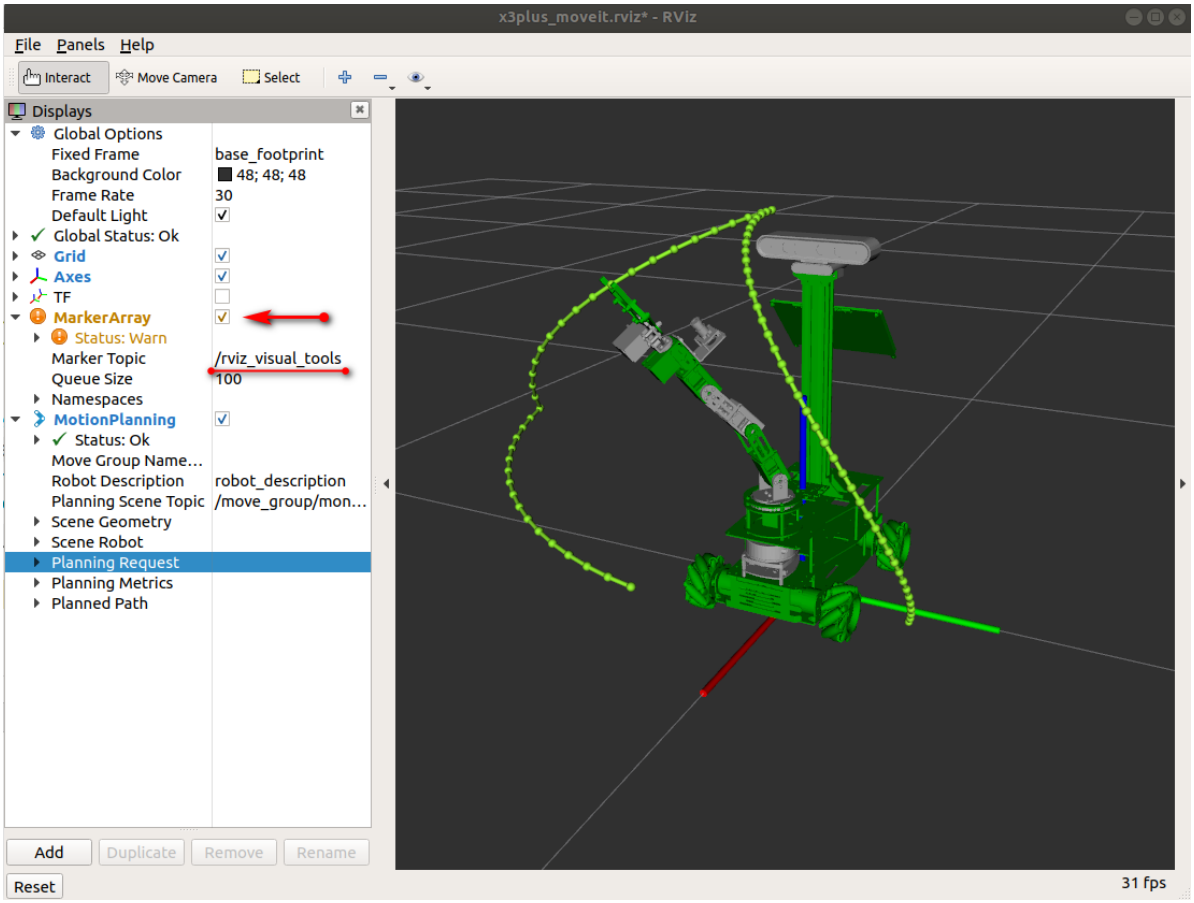
```
roslaunch arm_moveit_demo x3plus_moveit_demo.launch sim:=true
```

启动轨迹规划节点

```
rosrun arm_moveit_demo 06_multi_track_motion
```

效果图

查看轨迹需添加【MarkerArray】插件，选择【/rviz_visual_tools】话题。



给定机械臂三个可到达的目标点，MoveIT将根据目标点规划出三段可行轨迹，再将三段轨迹合并成一条连续轨迹。

8.2、源码解析

设置三个可达到目标点（目标点几个都可以，必须可到达）

```
    vector<vector<double>> poses{
            {1.34,  -1.0,  -0.61, 0.2,   0},
            {0,     0,     0,     0,     0},
            {-1.16, -0.97, -0.81, -0.79, 3.14}
    };
    for (int i = 0; i < poses.size(); ++i) {
        multi_trajectory(yahboomcar, poses.at(i), trajectory);
    }
```

规划每段轨迹

```
void multi_trajectory(
        moveit::planning_interface::MoveGroupInterface &yahboomcar,
        const vector<double> &pose, moveit_msgs::RobotTrajectory &trajectory) {
    moveit::planning_interface::MoveGroupInterface::Plan plan;
    const robot_state::JointModelGroup *joint_model_group;
    // 获取机器人的起始位置
    moveit::core::RobotStatePtr start_state(yahboomcar.getCurrentState());
    joint_model_group = start_state->getJointModelGroup(yahboomcar.getName());
    yahboomcar.setJointValueTarget(pose);
    yahboomcar.plan(plan);
    start_state->setJointGroupPositions(joint_model_group, pose);
    yahboomcar.setStartState(*start_state);
    trajectory.joint_trajectory.joint_names =
plan.trajectory_.joint_trajectory.joint_names;
    for (size_t j = 0; j < plan.trajectory_.joint_trajectory.points.size(); j++)
{

 trajectory.joint_trajectory.points.push_back(plan.trajectory_.joint_trajectory.
points[j]);
    }
}
```

轨迹合并

```
    moveit::planning_interface::MoveGroupInterface::Plan joinedPlan;
    robot_trajectory::RobotTrajectory rt(yahboomcar.getCurrentState()-
>getRobotModel(), "arm_group");
    rt.setRobotTrajectoryMsg(*yahboomcar.getCurrentState(), trajectory);
    trajectory_processing::IterativeParabolicTimeParameterization iptp;
    iptp.computeTimeStamps(rt, 1, 1);
    rt.getRobotTrajectoryMsg(trajectory);
    joinedPlan.trajectory_ = trajectory;
```

轨迹显示

```
    moveit_visual_tools::MoveItVisualTools tool(yahboomcar.getPlanningFrame());
    tool.deleteAllMarkers();
    /*
     ...
    */
    // 显示轨迹
    tool.publishTrajectoryLine(joinedPlan.trajectory_,
yahboomcar.getCurrentState()->getJointModelGroup("arm_group"));
    tool.trigger();
```

执行轨迹规划

```
    if (!yahboomcar.execute(joinedPlan)) {
        ROS_ERROR("Failed to execute plan");
        return false;
    }
```